🚦 ML-Based Intelligent Traffic Management System with Emergency Vehicle Priority Control

Overview 📌

This project is a Final Year Engineering Project that uses Machine Learning and Embedded Systems to optimize traffic signal control based on traffic density and provide priority to emergency vehicles such as ambulances using siren detection.

✨ Features

- Adaptive traffic signal control
- Emergency vehicle priority using siren detection
- Machine Learning based traffic classification
- Embedded hardware implementation using Arduino/ESP32
- Automated traffic signal switching
- Low-cost prototype for smart city applications

🛠 Tech Stack

- Python
- Machine Learning
- Arduino / ESP32
- Embedded C
- Sensors
- Basic IoT

⚙ Hardware Components

- Arduino / ESP32
- Sound Sensor / Microphone
- LEDs (Traffic Signal)
- Breadboard

- Jumper Wires
- Power Supply

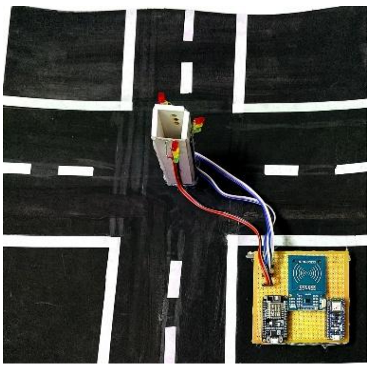
📂 Project Structure

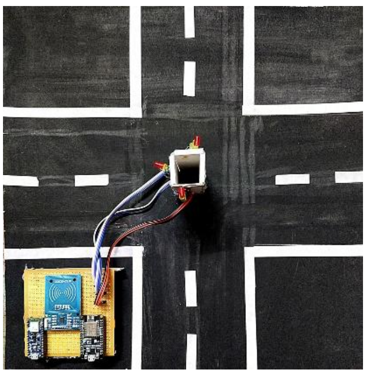
```
project/
├── README.md
├── code/
├── hardware/
├── images/
├── report/
└── presentation/
```

🚀 Working

1. Detect traffic density.
2. Predict traffic condition using Machine Learning.
3. Detect ambulance siren.
4. Override normal traffic signal.
5. Give green signal to emergency vehicle.
6. Resume normal traffic operation.

📸 Prototype





📊 Future Improvements

- Computer Vision using CCTV

- Cloud Dashboard

- GPS-based emergency vehicle tracking

- Real-time traffic analytics

👩‍💻 Author

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